Indoor Localization using RSSI based Supervised Machine Learning Approaches

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***Abstract*—Wireless indoor localization has attracted significant interest from industrial and academic domains due to its crucial role in location-based applications. However, localization is not an inherent feature of the ZigBee devices; it is possible to employ wireless technology capabilities for localization. This paper uses machine learning models to map the indoor localization system between the received signal strength indicator (RSSI) and the node position. A system map is created using a fingerprint technique containing an RSSI (input)position(output) pair. To study the influence of RSSI on the localization algorithm, we proposed framework employs four different Supervised Machine Learning models, including K-Nearest Neighbour (KNN), Feed-Forward Neural Network (FF), Cascade-Forward Neural Network (CF), and Long-Short - Term Memory-Recurrent Neural Network (LSTM-RNN). The results showed that Wi-Fi technology achieved the best performance in the CF model over all.**

Keywords—Supervised Machine Learning, RSSI, Localization, Fingerprint, ZigBee.

I. Introduction

The increasing number of wirelessly connected devices accompanies the increasing necessity of the demand for location. Applications such as smart homes, self-navigating robots, industry, surveillance, monitoring, building energy conservation [1], and medical applications require an accurate location. Localization estimates the node devices' coordinates (smartphone, sensor, or any device with a wireless connection).

Localization is divided into two kinds: outdoor and indoor. In the outdoors, the Global Positioning System (GPS) is an optimum standard solution already adopted to answer positioning problems with an accuracy of around 3.5 meters [2]. This solution is improper for indoor localization due to different limits, such as line-of-sight (LoS) necessity, high cost, energy consumption requirement, and low accuracy.

Several kinds of research have been done to fulfill indoor localization using Bluetooth, ultrasonic, infrared, ZigBee, and Wi-Fi [2]. According to network infrastructure (isotropic and anisotropic), the localization techniques are range-free and range-based [3]. In range-free localization(isotropic), the corresponding distance between the nodes is estimated according to the hop counting. This method is adopted in some literature, but the range-based (anisotropic) one is most real because of the nature of the indoor environment[4]. In range-based localization, ranging techniques are used to perform distance measurements to estimate the locations of the nodes with initially unknown positions. These measurements use the variable characteristics of electromagnetic waves to measure the distance between unknown and reference or anchor nodes[5].

Several localization methods were proposed, the most public of which is the Receiving Signal Strength Indicator (RSSI) [6], which measures the decay in signal power depending on the distance. This method is more appropriate for localization because it does not need synchronization or additional devices.

Two methods can be used to estimate the position of unknown node devices using RSSI: path-losses-based model and fingerprint-based. The path-loss model approach estimates the distance between two-node devices using RSSI [8]. This distance cannot be calculated accurately unless the exact pathlosses model of indoor environment characteristics is a priori known. However, this cannot be calculated in general form because of many obstacles that lead to errors produced by building that may affect communication channels. The path losses model-based Localization systems show non-negligible localization errors due to the high fluctuations in RSSI measurements. The other method is the fingerprint-based localization method, which has better accuracy and precision than the path-losses model-based [9]. The fingerprint method procedure comprises two steps: infra-data collection and real-time estimation. In the former step, a fingerprint map is constructed by dividing the area into small pieces with a known coordinate and reading the RSSI value captured from several access points (APs) in each coordinate to create a reference point (RP). In the later step, the real position of the node device is estimated using the RP and localization algorithm (LA).

LA can be traditional by converting the RSSI value to distance or by a more developed method. Thus, the effects were affirmed before hindering the traditional techniques, such as trilateration or angulation, to perform an accurate

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location estimation[10]. These traditional techniques must be replaced with more advanced ones to overcome these problems, such as Supervised Machine Learning (SML). The quick learning capability, flexible modeling and ease of use make SML one of the best candidates for localization in dynamic wireless environments [11]. In addition, this method does not need to know a priori the characteristics of wireless channels. Instead of distance estimation, the SML architectures directly estimate the unknown target location[12] using a similarity metric to estimate the online data based on fingerprint data. Then the target is positioned at the nearest fingerprint location in the signal space[13]. Despite its highest accuracy[14], the SML is sophisticated and needs high computational complexity in the training and inference phase[15]. In this paper, we proposed comparing four SML types: KNN, CC, CF, and LSTM-RNN.

The rest of this paper is divided into sections: Section 2 contains the related work. Section 3 contains the proposed framework, and section 4 contains the experimental results and discussion.

II. RELATED WORKS

Recently, a vast amount of study has been accomplished to perform localization operations inside a building. Haibing Hu et al. [16] collected 25 fingerprints from four ZigBee module CC2530 that were placed in the corner of an 8m x 8m empty area and used a Kalman average filter to estimate the distance from point-to-point RSSI and rebuild the RSSI according to the estimated distance. However, the obtained dataset is used to train the ELMAN neural network, which is used for target coordinates estimation. The localization accuracy achieved by this method was 0.915 meters.

Aiman Ibrahim et al. [17] proposed evaluating the performance of three neural network algorithms: Feedforward Backpropagation Neural Network, Cascade-forward Backpropagation Neural Network, and Elman Backpropagation Neural Network. Simulated RSS are generated based on a path loss model with a distance of up to 100 meters. The results show that the best performance was achieved with the ELMAN network, which has an MSE of 0.4991m. Sadowski et al. [18] proposed indoor localization based on Wi-Fi, BLE, and ZigBee technologies. These technologies propagate their signal in three experimental scenarios: low, moderate, and high interference. Three localization algorithms are evaluated: KNN, Naive Bayes, and trilateration. Regarding the technologies, the authors declare that "the indoor localization system is not affected significantly by the utilized wireless technology." Regarding the algorithm, the best overall performance was achieved with KNN. Bing Jia et al. [19] proposed a long short-term memory network (LSTM) algorithm to perform indoor localization, and the authors sprinkled samples using Gaussian process regression and principal component analysis. The dataset used in this study was UJIIndoorLoc, a Wi-Fi fingerprint collected from multi buildings and floors in a California university. The results show that the methods used outperform the other methods that used this dataset. This paper divides the work into two phases to trade-off between accuracy, precision, installation time, and labor. An RSSI fingerprint was initially provided in three different environments from three well known wireless technologies (Wi-Fi, ZigBee, and Bluetooth low energy (BLE)). The fingerprint was utilized to train five supervised machine learning. As the target is localization, we compare based on accuracy and precision. In the second phase, the fingerprint is generated using a real measured dataset by considering only the maximum and minimum RSSI and examining the system's performance.

III. THE PROPOSED MAPPING FRAMEWORK

This section will describe the mapping framework in detail, including the proposed technique, machine learning models, wireless technologies, and datasets.

A. Fingerprinting

The fundamental principle of fingerprinting is that every predefined coordinate at the site of interest possesses a unique fingerprint (FP). An FP consists of measurements derived from radio signals (such as CSI and RSSI) emitted by nearby beacon nodes, magnetic fields, and other location-related information [20]. The Fingerprinting technique is divided into two parts: offline and online. The offline part prepares the database for the subsequent online part. The procedure of offline starts by dividing the location of interest into multi-piece, each piece identified as a Reference Point (RP). Subsequently, signals from distributed beacon nodes or associated sensors are sampled numerous times at each RP to generate a fingerprint. Then, all fingerprints and the position coordinates of their related RPs are stored in a database. In the online part, real-time measurements by localization techniques match existing fingerprints in the database. The estimated position is the coordinates of the RP whose fingerprint produces the closest match [21]. K-nearest neighbor (KNN) and artificial neural network (ANN) are typical localization strategies. In addition to utilizing the correct matching procedure, the estimation precision is enhanced by distributing additional RPs around the location of interest. The more RPs used; the smaller region designated by each RP becomes. However, the problem is that the measurements taken at various RPs may be highly close, making their fingerprints challenging to differentiate.

B. Localization techniques

All localization techniques used in this paper are supervised learning. The learning algorithm predicts unknown data in supervised learning depending on the previously saved datasets. For the localization step, a labeled sample dataset exists (coordinate labeled by RSSI group). A supervised learning algorithm aims to build a mapping function (model) to predict a given RSSI input's coordinate (x, y). This category of algorithms is known as supervised learning since the correct label for every feature vector is known beforehand[22]. The choice of localization techniques was deliberate and methodical. We selected KNN because it is a deterministic method known for its high accuracy and precision when applied to this dataset [22]. Additionally, the selection of the other techniques was guided by two specific criteria, as detailed below:

- The contribution of RSSI.

Two techniques were selected for studying the contribution of RSSI in the network a) Feedforward neural network, where the RSSI contributed in the input layer only. b) Cascade forward network, where the RSSI contributed in each layer.

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- The data is sequentially affected.

The LSTM-RNN is work with sequential data, where the last RSSI sequence contributes to the recent decision. This section explains each technique used in detail.

For this work, the best FF and CF performance is obtained with two hidden layers and twelve neurons for each layer, while the best results are obtained for LSTM-RNN with one layer and twenty neurons.

1) *K-Nearest Neighbour (KNN)*

KNN algorithm is instant-based learning adopted for addressing the regression and classification problem [23][24]. The KNN algorithm calculates distances between measured RSSI values from the beacon and RSSI of the examples in the dataset using the Euclidean Distance, as follows:

$$\mathrm{Di} = \sqrt{\sum_{j}^{n}\left(RSSI_{ij} - RSSI_{j}\right)^{2}} \quad ,\mathrm{i} = 1,2,3,\ldots \mathrm{M} \qquad (1)$$

where, Di represents the space measured between the $RSSI_j$ at beacon j of a test location and the saved fingerprint $RSSI_{ij}$ at location i; n denotes of the beacon number in the database and M is the entry number in the database.

After calculating the distance between all points in the database, the k closest matches are picked, and their (x,y) coordinates are averaged to derive a final estimate of the receiver's location according to the following equation[23] :

$$(\mathrm{x},\mathrm{y}) = \frac{1}{k}\sum_{i=1}^{k}(x_i, y_i) \qquad (2)$$

The best performance of KNN in this work was obtained with $k = 4$.

2) *Feed-forward backpropagation network (FFNN)*

The main structure of this type of neural network includes three layers an input layer, a hidden layer, and an output layer. The input layer receives data, normalizes them, and distributes them to the hidden layer. The hidden layer calculates and transfers the results to the output layer, which normalizes and extracts final results. In a feed-forward neural network, the outputs of each layer's neurons are fully connected to all the neurons in the subsequence layer, and all the layers contribute to making the final decision without receiving any feedback. This kind of neural network is commonly employed in fingerprint-based localization. Equation 3 is an example of inference equation of two hidden layers [25]:

$$(\mathrm{x},\mathrm{y}) = AF_{out}(AF_2(AF_1(RSSI_n.W_{i1} + b_{i1}).W_{2j} + b_{2j}).W_{3k} + b_{3k} \qquad (3)$$

Where, AF is the activation function. $RSSI_n$ is the vector of RSSI acquired from the access point. W denotes weight vector. b is the bias vector.

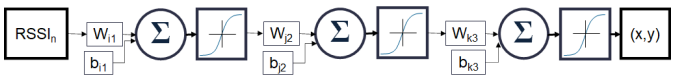

Fig. 1 Structure of feed-forward back propagation network

3) *Cascade-forward back propagation network (CF)*

Like feed-forward networks, the CF model incorporates a weighted link from the input to each layer and from each layer to the following levels, as illustrated in figure 2; While CF with two layers can learn almost any input-output link, Cascade forward networks with more layers may predict complex correlations more quickly. Equation 4 describes the mathematical route of the inference phase for two hidden layers cascade neural network.

$$(\mathrm{x},\mathrm{y}) = AF_{out}(AF_2(AF_1(RSSI_n.W_{i1} + b_{i1}).W_{2j} + b_{2j} + RSSI_n.W_{i2}) .W_{3k} + b_{3k} + RSSI_n.W_{i2} + (AF_1(RSSI_n.W_{i1} + b_{i1}).W_{j2}) \qquad (4)$$

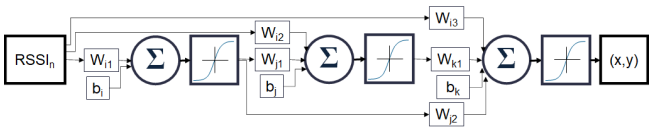

Fig. 2 Structure of cascade- forward back propagation network

4) *Long Short-Term Memory — Recurrent Neural Network (LSTM-RNN)*

LSTM is one of the recurrent neural networks (RNN) types invented to model the data with time sequences. LSTM is more accurate than conventional RNNs due to its long-range temporal dependency. In conventional RNN architecture, the backpropagation algorithm causes a back-flow error. LSTM overcomes this problem by adding special units in the recurrent hidden layer called Memory Blocks (MBs) [26]. MBs contain memory cells with self-connections and three types of gates. Memory cells' store the temporal status of the network. The gates are responsible for controlling the flow of information[27]. The gates are named depending on their position as described in the following:

- The input gate is responsible for the flow of input activations into the memory cell controls.
- Output gate manages the output cell activation's flow to other network parts.
- The forget gate scales the cell's internal status before summing it as input to the cell through its self-recurrent connection, thus resetting the cell's memory adaptively. In addition, the updated LSTM architecture can learn the accurate output timing by helping the peephole that connects the internal cells with the gates in the same cell.

C. *Datasets*

The RSSI dataset used in this work is taken from [28], which was collected from an empty small meeting room out of interference of 33 (6 x 5.5) square meters area as seen in fig. 2. The authors use three access points (AP) to build a test bench in the three room angles. From this test bench, 49 fingerprint reference points (RP) and 10 test points (TP) are read as mentioned in fig.2. The distance between these reference points is 0.5 meters.

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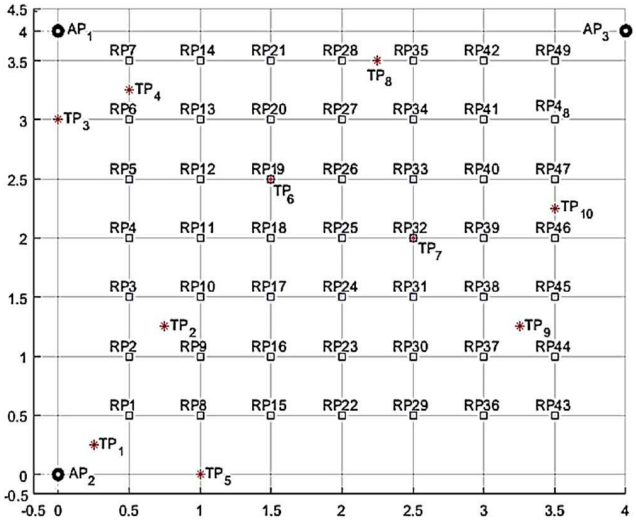

Fig. 4 testbench small room (6 x 5.5 m)

D. Zigbee wireless technology

A wireless technology must be chosen when designing an indoor localization system, which is a crucial consideration. Numerous different types of localization systems have been developed using technologies like Wi-Fi [31], Bluetooth Low Energy (BLE), ZigBee, Radio Frequency Identification Device (RFID), and Ultra-Wideband (UWB); thus, there is no industry standard in place, so designers must choose the wireless technology based on the needs of the application [9]. In this paper we select the Zigbee technology due its low power comparing with Wi-Fi and high-range comparing with Bluetooth. ZigBee's communication protocol is renowned for its ease of use, low power consumption, and secure networking features. The IEEE 802.15.4 standard, which specifies the operational point for wireless personal area networks (WPANs) with low-data rate antennas, is the foundation of the ZigBee protocol. By employing carrier-sense multiple access with collision avoidance (CSMA/CA), devices using IEEE802.15.4 can regulate the information flow and avoid data loss [33]. ZigBee-enabled devices are built with attributes like link quality and energy sensing, making assessing measures like the RSSI simple. Compared to BLE, ZigBee has a broader range because it can broadcast data farther, thanks to a mesh network of relay nodes. The widespread adoption of ZigBee for WSN localization results from its low power needs, but since additional hardware is needed, it is less common among IoT users today

IV. Experimental Results and Discussion

A. Experimental setting

The environmental scenarios are explained in detail in the previous section real dataset. Concerning the system performance analysis, the performance criterion measures the ability of localization techniques to predict the unknown position with minimum distance error. Two criteria can be used to determine the accuracy of the system[35]:

RMSE measures the root of summing the square difference between the coordinate of actual data and estimated data as illustrated in the equation below:

$$\text{RMSE} = \sqrt{(x_{actual} - x_{measure})^2 + (y_{actual} - y_{measure})^2} \quad (7)$$

Variance is calculating the differences between each number in the data set and the mean, then squaring the differences to make them positive, and finally dividing the sum of the squares by the number of values in the dataset as illustrated in equation 8 below:

$$\sigma^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})}{n-1} \quad (8)$$

where, x_i is i_{th} data point. $\bar{x}$ denotes Mean of all data points, and n is number of data points.

A minor variance indicates that numbers in the set are near each other's.

B) Results

We simulate the dataset for the testbench mentioned above with four localization techniques. Figures 5 show the relation of cumulative distribution function (CDF) for the proposed testbench, concerning the distance error for different techniques. Table 1 summarizes the variance and mean square error calculation for three scenarios. We will explain the scenario results in detail by starting with CDF and finishing with the Table 1 results.

For the CDF results are shown in Figure 5; in the full-time, the LSTM-RNN has the best performance with an error of 2.7m. KNN followed it with an error of 3 m, CF with an error of 3.1m, and FF with an error of 3.5. In 90% of the time, the best performance was for CF with an error of 2.4m, followed by KNN with 2.5m, and FF with an error of 3m.

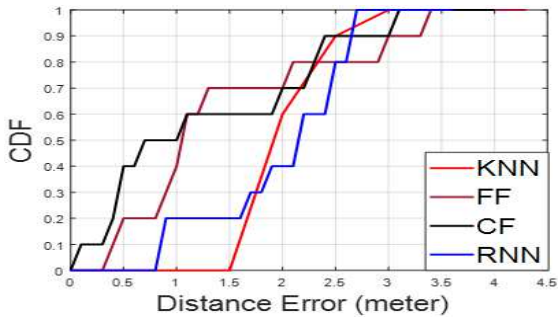

Fig. 5 CDF response for out of interference scenario ZigBee.

In the Table 1, the best algorithm is CF, with localization error of 1.2164 m and variance of 1.0600 m, followed by FF with an error of 1.3324m, FF with an error of 1.4491m and variance of 1.0368. the third position was for LSTM-RNN with localization error of 1.6025 and variance of 0.4298 and at the end, KNN with the error of 1.8376m.

TABLE I. Summary of Error and Variance for Localization Techniques

Localization techniques	MSE	σ^2
KNN	2.0007	0.1672
FF	1.4491	1.0368
CF	1.2164	1.0600
LSTM-RNN	1.6025	0.4298

(7)

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C) Discussion

The results obtained with the algorithms trained by the dataset declared that the overall results show that the best algorithm is CF. All algorithms were trained three times with different layers, neurons, and activation functions. The algorithm with low RMS error is selected except the KNN trained with each run and has fixed results. The results indicate that all the algorithms can perform indoor localization. It must be noted that, unlike the other algorithm, the advantage of KNN is that the algorithm will consider any change in the dataset due to its instant training with each estimation. In contrast, any change in the database when using FF, CF, and LSTM-RNN requires retraining the algorithms.

V. Conclusions

This work used supervised machine learning to perform indoor localization tasks. We evaluated four localization techniques on dataset. The dataset selected was built by ZigBee wireless technology in low interference scenario. The techniques are compared in terms of accuracy and precision. The Cascade Forward neural network (CF) appears to have the best behavior, followed by Feed Forward Neural Network (FF) and Long Short-Term Memory (one LSTM with 30 neurons) come in the third position, and the last is K Nearest Neighbor with k=4. The CF, and FF with two hidden layers of twelve and fifteen neurons. The proposed frameworks can perform an indoor localization system with low localization error.

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